

Over-refusal direction analysis (partial). The harmful-refusal sample count in Qwen is $n=1$, leaving the DIM harmful-refusal direction undefined and making the central directional comparison ($\cos(\hat{\mathbf{v}}_\ell^{\text{OR}}, \hat{\mathbf{r}}_\ell)$) impossible. Nevertheless, we characterise the over-refusal direction itself. We compute the global over-refusal DIM direction $\hat{\mathbf{v}}_\ell^{\text{OR}}$ at each layer as the difference-in-means between over-refusal ($n=22$) and target-answered samples, and track its L2 magnitude across all 31 layers (Figure 16, left). The magnitude profile peaks in early-to-mid layers near the constellation peak at layer 5, consistent with the LLaMA finding that the OR direction is most pronounced at the peak task-cluster layer.

Per-task OR direction analysis is limited to a single task (cryptanalysis, the only task with sufficient over-refusal samples in Qwen), so inter-task pairwise cosine similarity cannot be computed. We instead track the layer-to-layer angular stability of the global over-refusal direction (Figure 16, right): the consecutive-layer cosine rises through mid-layers as task structure crystallises and then stabilises. Full cross-task directional comparison requires an evaluation set that elicits over-refusal across at least two tasks in Qwen.

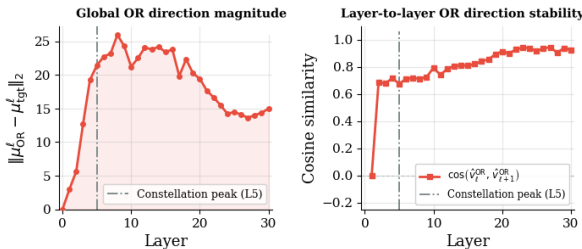


Figure 16: Qwen1.5-7B: global OR direction magnitude across layers (left) and layer-to-layer angular stability (right). The harmful-refusal DIM direction is unavailable.

Over-refusal subspace dimensionality (partial).

We compute the principal-component structure of the Qwen over-refusal subspace at layer 5 by centering the OR embeddings on the target-answered mean and running PCA (Figure 17). The first principal component explains 64.1% of the variance ($n_{80}=2$), far more concentrated than the LLaMA over-refusal subspace at its peak layer ($\text{PC1} \approx 24.5\%$, $n_{80}=11$). This apparent concentration is a single-task artefact: Qwen’s 22 over-refusal samples arise predominantly within the cryptanalysis task, so the subspace collapses onto

the principal axis of that one cluster. When over-refusal spans multiple tasks—as in LLaMA, where sentiment analysis and translation both contribute—the subspace spreads across task-specific axes and becomes genuinely higher-dimensional. The Qwen result therefore does not contradict Claim 3; it instead illustrates the mechanism: a single-task OR population produces a compact subspace precisely because it lacks the cross-task spread that higher dimensionality requires. The harmful-refusal subspace cannot be computed ($n_{\text{RH}}=1$), so the OR vs. RH dimensionality comparison is unavailable.

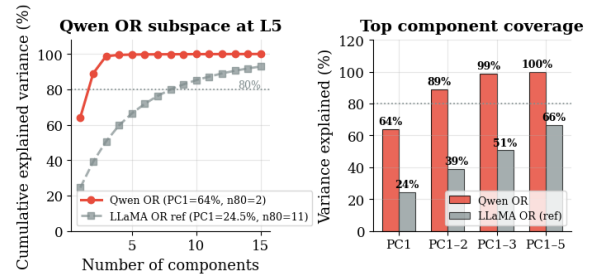


Figure 17: Qwen1.5-7B OR subspace at layer 5 (cumulative variance left; top- k coverage right) versus LLaMA OR reference. Qwen OR is highly concentrated ($\text{PC1} \approx 64\%$) because over-refusal samples span only a single task (cryptanalysis).

Linear probing. We apply the same logistic-regression probing pipeline as in §4.3 to Qwen residual-stream activations, independently for task identity (5 tasks, $n=270$) and refusal behaviour (over-refusal vs. target-answered; Figure 18). The task-identity probe achieves high balanced accuracy from the first layer and saturates before the constellation peak at layer 5, replicating the LLaMA finding that task identity is encoded early. The refusal-behaviour probe rises through mid-layers and peaks after the task-identity probe has saturated, consistent with over-refusal being a within-task perturbation layered on top of established task structure. The refusal-type probe (over-refusal vs. harmful-refusal) cannot be run: with $n_{\text{RH}}=1$, five-fold cross-validation is infeasible. The qualitative ordering—task identity decodes before refusal behaviour—mirrors LLaMA (task probe peak 0.986 at L6; behaviour probe peak 0.939 at L15) and supports the interpretation that this representational hierarchy is consistent across model families.

The cross-model replication contributes a model-independent lower bound: task-identity structure of comparable strength (silhouette ≈ 0.5) forms in

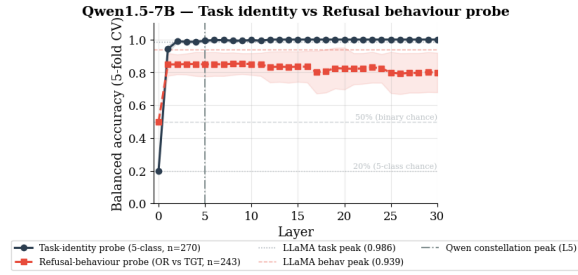


Figure 18: Qwen1.5-7B: linear probe accuracy across layers for task identity (black) and refusal behaviour (red). Task probe saturates closely with behaviour probe.

both architectures, confirming that the geometric substrate for over-refusal is not specific to LLaMA-3.1. In Qwen, the task-identity constellations replicates fully. The analysis on directional geometry and OR vs. RH dimensionality is limited by Qwen’s near-zero harmful-refusal rate and hence, OR direction magnitude and OR subspace are partially characterised and their qualitative behaviour is consistent with LLaMA.